Moving data

Before packets can be sent between devices, they have to be translated into binary signals (ones and zeroes) that can travel over great distances. Every device on the Internet has a "network adapter" to perform this task. Different devices send data in different forms.

△ **Electrical signals** Copper wires carry ones and zeroes as electrical signals of different strengths.

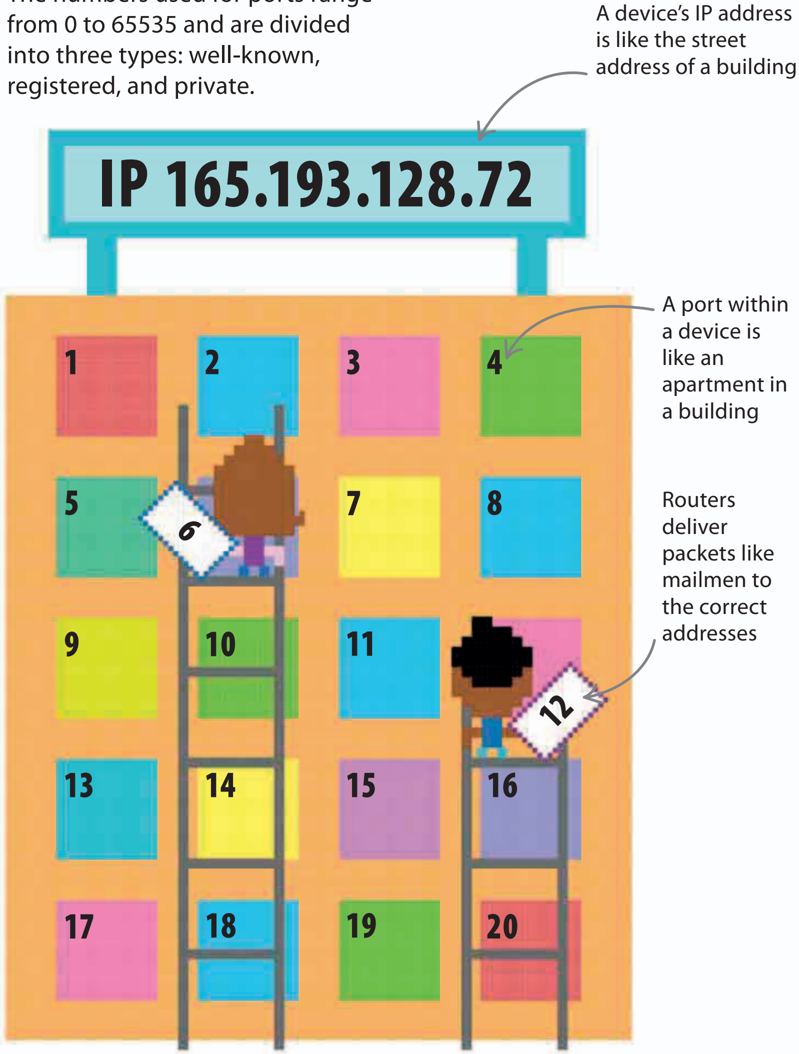
△ **Light** Special glass fibres, called fibre optic cables, transmit data as pulses of light.

△ **Radio waves** Different types of radio waves can carry ones and zeroes without using wires.

Ports

Just as you mail a letter to a specific person in an apartment building, you may want to send packets to a specific program on a device. Computers use numbers called "ports" as addresses for individual programs. Some common programs have ports specially reserved for them. For example, web browsers always receive packets through port number 80.

▽ **Port numbers** The numbers used for ports range from 0 to 65535 and are divided into three types: well-known, registered, and private.



EXPERT TIPS

Sockets

The combination of an IP address and a port is known as a "socket". Sockets let programs send data directly to each other across the Internet, which is useful for things such as playing online games.